Numerical Analysis of Quasioptical Multireflector Antennas in 2-D With the Method of Discrete Singularities: E-Wave Case

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***Abstract*—Numerical method for modeling the E-polarized wave scattering by electrically large quasioptical two-dimensional (2-D) reflectors is presented. Reflectors are assumed zero-thickness and perfectly electrically conducting. Efficient numerical solution is obtained from the coupled singular integral equations discretized using new quadrature formulas of interpolation type. It has controlled accuracy and deals with small-size matrices. To simulate a small-horn feeding, the incident field is taken as a beam generated by a complex-source-point (CSP) current. Presented numerical results validate empirical rule of -10 dB edge illumination needed to provide the best electromagnetic performance of reflector.**

***Index Terms*—Boundary integral equations, method of discrete singularities (MDS), reflector antennas, scattering.**

I. Introduction

CURVED large-size metallic reflectors are used today in many applications from the cheap parabolic dishes for satellite TV receivers to the systems of deep space communication with Mars orbiters and rovers. Satisfactory analysis and even synthesis of reflectors is usually done with optical and "quasioptical" methods, namely geometrical optics and physical optics (GO and PO), supplemented with geometrical theory of diffraction and the uniform theory of diffraction (GTD and UTD) and Gaussian-beam field decomposition [1]–[8]. Being very powerful tools, these methods still do not characterize fine wave effects in complete manner. This may lead to inefficient performance of geometrically designed multireflector systems known, e.g., as "low-frequency problems" [9]. For a more accurate analysis, full-wave methods are mandatory. At the same time, popular today FDTD field solvers require prohibitively large computer resources when applied to reflector antennas.

A general way to build more economic full-wave tools for the modeling of reflector-type scatterers is to use integral-equation (IE) approach [10]. Here, the crucial point is development of efficient discrete model, i.e., fast and convergent numerical algorithm having controlled accuracy.

Although sometimes vulnerable to poor convergence [11], method-of-moments (MoM) with local basis functions gives satisfactory results for small and medium size reflectors [12]. However, it fails if they are much larger than the wavelength λ - see remarkable report [13]. The main conclusion of [13] is worth citation "a theoretically exact method (such as MoM) cannot be practically used for antennas larger than a few λ." Note that somewhat earlier it was stated [14] that the use of global basis functions widened the range of computable reflectors to 25λ.

Today, it is commonly believed that the domains of acceptable accuracy of GO/PO and MoM overlap on the 5-λ to 20-reflectors [12], [14]. Although reasonable for single reflectors, this is not so obvious for the multireflector and reflector-in-radome systems, especially in the millimeter and submillimeter ranges characterized with extended near zones. As a remedy, discrete models based on the method of analytical regularization (MAR) or analytical preconditioning, i.e., conversion of a singular IE (SIE) for the 2-D surface current function to a Fredholm second kind matrix equation were proposed in the 1990s [15], [16]. We present here the basic ideas of another efficient discrete model based on the method of discrete singularities (MDS) [17]–[19], and demonstrate its potentials in the accurate modeling of electrically large quasioptical multireflector antennas in 2-D.

MDS is a Nystrom-type numerical technique adapted to the open strip-like scatterers with edges and hence its success depends on finding the appropriate quadrature rule. A similar method, of the vortex lattices, was used earlier in fluid dynamics [20]. After the original submission of this paper, Nystrom-type solutions to flat-strip scattering problems were published in [21] and [22]. Note, however, that in [21] the strip contour is divided to large number of segments, where specific edge-corrected interpolation scheme is applied only to the edge-adjacent segments. In contrast, here we develop a Nystrom-type algorithm based on the quadrature rules that account for both the singularity of the kernels and the singularities of solutions at the edges, applied to the whole reflector, i.e., we do not discretize its contour into segments. This is the same idea as in [22] however we, first, use it for curved contours, and, second, we differentiate our SIEs and then deal with the Cauchy singularities in kernels instead of logarithmic ones as in [22].

In the remainder of the paper, Section II presents basic equations of the MDS in the case of the E-polarization. In Section III we numerically analyze various single reflectors fed by the E-polarized beam-like fields. In Section IV, we extend this analysis to multireflector antennas to demonstrate the capabilities of MDS. Conclusions are summarized in Section V. Time dependence is given by $e^{-j\omega t}$ and omitted.

Manuscript received November 7, 2005; revised August 31, 2006. The work of A. A. Nosich was supported by undergraduate and graduate student awards from the IEEE Antennas and Propagation Society, the IEEE Microwave Theory and Techniques Society, and the SUMMA Foundation.

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Digital Object Identifier 10.1109/TAP.2006.889811

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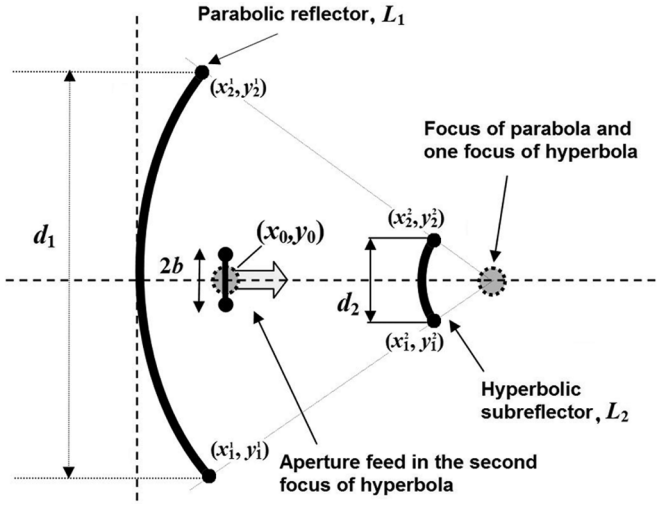

Fig. 1. 2-D Geometry of symmetric Cassegrain antenna front-fed by an aperture source-an example of a two-reflector scattering structure.

II. Basic Concepts of MDS

A. Problem Formulation

We study the scattering of the E-polarized waves by $Q \geq 1$ finite-size cylindrical reflectors, which model a quasioptical beam-forming system. The geometry of a generic two-reflector structure and an aperture feed is shown in Fig. 1. As perfectly electrically conducting (PEC) reflectors are supposed to have zero thickness, we assume that $L_q = \{(x_q, y_q) \in R^2 : x_q = F_q(y_q), y_2^q < y_q < y_2^q\}$ are smooth nonintersecting curves, $q = 1, 2, \ldots Q$.

The total field in the presence of reflectors is a sum, $U^{\text{tot}} = U_0 + U \equiv U_0 + \sum_{q=1}^{Q} U_q$, where U_0 is the given incident field, and U_q is the secondary field scattered by the q-th reflector. Electromagnetic boundary-value problem requests U to solve the Helmholtz equation of $L = \bigcup_{q=1}^{Q} L_q$

$$(\Delta + k^2)U(x, y) = 0, \quad (x, y) \in R^2 \backslash L \tag{1}$$

where $k = \omega/c = 2\pi/\lambda$ is the wavenumber and c is the light velocity, and satisfy the PEC boundary condition on L, i.e., the Dirichlet condition in the case of the E-polarization

$$U(x, y)|_L = -U_0(x, y)|_L. \tag{2}$$

Besides, U must satisfy Sommerfeld radiation condition

$$\partial U(x, y)/\partial r - jkU(x, y) = \bar{o}(r^{-1/2}), \quad r \to \infty \tag{3}$$

and condition of the local energy finiteness

$$\int_{\Omega} (k^2|U|^2 + |\nabla U|^2) d\sigma < \infty \tag{4}$$

where Ω is a bounded domain in R^2 containing endpoints off L. These problems are uniquely solvable [10].

B. Source Modeling

In quasioptical antenna modeling, primary feed field should have directive beam-like form that simulates the radiation of a small horn. Conventional line current is not a good choice for this. If, however, a line current is placed at the complex-valued source point, its field becomes

$$\begin{aligned} U_0(\vec{r}) &= H_0^{(1)}(k|\vec{r} - \vec{r}_c|), \quad \vec{r} = (x, y), \\ \vec{r}_c &= \vec{r}_0 + j\vec{b}, \quad \vec{r}_0 = (x_0, y_0), \vec{b} = (b\cos\beta, \quad b\sin\beta) \end{aligned} \tag{5}$$

where $H_0^{(1)}(z)$ is the Hankel function of the first kind and zero order, and other notations are shown in Fig. 1. Function (5) is a rigorous solution to the Helmholtz equation. For all $|\vec{r} - \vec{r}_0|$ it displays maximum intensity in the direction $\phi = \beta$. It behaves as a Gaussian beam in the near zone of paraxial domain and transforms smoothly into a cylindrical wave in the far zone—see [1], [15], and [23]. Mathematically, function (5) has two branch points at $(x_0 \pm b\cos\beta, y_0 \mp b\sin\beta)$, and thus to single out a unique value of $U_0(x, y)$ one has to join them with a branch cut B of the length 2 b in the real space. This cut can be considered as a model of the aperture of a real horn feed, whose radiation field is simulated by the CSP beam field field. Note that the greater kb, the narrower the CSP beam and the greater the distance of significant field intensity. In our simulation, the cut B should not intersect the contours of reflectors L.

C. SIE for the E-Polarization Case

Potential theory reduces the problem (1)–(4) to a set of Q logarithmic-type electric-field SIE of the first kind for the electric currents induced on the reflectors, i.e., $j_q\ (s)$

$$\begin{aligned} \sum_{q=1}^{Q} \int_{L_q} H_0^{(1)}(k|\vec{r}_p(s_q) - \vec{r}_q(s_{0_q})|) j_q(s_q)\, ds_q \\ = -U_0(\vec{r}_p(s_{0_p})), \quad p = 1, 2, \ldots Q \end{aligned} \tag{6}$$

with s_q for the arc length along the curve L_q.

Now, introduce C^2-smooth parameterizations $x_q(t)$, $y_q(t)$ to L_q, where $t = 2s/|L_q| - 1$, and $|L_q| = \int_{-1}^{1} [\dot{x}_q^2(t) + \dot{y}_q^2(t)]^{1/2} dt$ is the length of L_q, hence $|t| \leqslant 1$. Recalling singular behavior of the unknown current functions $j_q\ (s)$ at the endpoints of L_q, we substitute them for new and smooth ones according to the identities $j_q(t)[(\dot{x}_q(t))^2 + (\dot{y}_q(t))^2]^{1/2} = v_q(t)(1 - t^2)^{-1/2}$. They are to be determined from a set of canonical SIEs as follows:

$$\sum_{q=1}^{Q} \int_{-1}^{1} H_0^{(1)}\left(kR_{pq}(t, t_0)\right) v_q(t) \frac{dt}{(1 - t^2)^{1/2}} = -U_0(t_0) \tag{7}$$

where $R_{pq}(t, t_0) = [(x_q(t) - x_p(t_0))^2 + (y_q(t) - y_p(t_0))^2]^{1/2}$.

Direct discretization of (7) is not efficient due to the logarithmic singularities of the kernel functions; instead, MDS suggests a transformation to another SIE set with stronger singularities. Such transformation to a set of Cauchy-type SIEs with supplementary conditions is done by: i) differentiation of (7) in t_0 and ii) integration of (7) on the $[-1, 1]$ interval, in t_0, with the

weight function $(1-t_0^2)^{-1/2}$ [17]–[19]. As a result we obtain a set of equations

$$\sum_{q=1}^{Q}\left[\delta_{pq}\int_{-1}^{1}\frac{v_q(t)}{t-t_0}\frac{dt}{(1-t^2)^{1/2}} + \int_{-1}^{1}K_{pq}(t,t_0)\frac{v_q(t)dt}{(1-t^2)^{1/2}}\right]=f_p(t_0)$$

$$\sum_{q=1}^{Q}\left[\int_{-1}^{1}M_{pq}(t)v_q(t)\frac{dt}{(1-t^2)^{1/2}}\right]=c_p,\quad p=1,2,\ldots Q. \quad (8)$$

Here, $f_p(t_0) = (-U_{p,0}(t_0))', c_p = -\int_{-1}^{1}U_{p,0}(t_0)(1-t_0^2)^{-1/2}\,dt_0$ is known constant, and K_{pq} and M_{pq} are known smooth functions

$$K_{pq}(t,t_0)=\frac{\partial}{\partial t_0}H_0^{(1)}(kR_{pq}(t,t_0))-\frac{\delta_{pq}}{t-t_0} \quad (9)$$

$$M_{pq}(t)=\int_{-1}^{1}H_0^{(1)}(kR_{pq}(t,t_0))\frac{dt_0}{(1-t_0^2)^{1/2}}. \quad (10)$$

D. Discretization of SIEs With MDS

Cauchy-SIE (8) is further discretized by using the quadrature formulas of interpolation type (see [17]) with the nodes taken at the following points: $t_i^n=\cos[(2i-1)\pi/(2n)], i=1,2,\ldots n$, i.e., the nulls of the Chebyshev polynomials of the first kind, $T_n(t_i^n)=0$, and $t_{0j}^n=\cos(j\pi/n), j=1,2,\ldots n-1$, i.e., the nulls of similar polynomials of the second kind, $U_{n-1}(t_{\mathrm{oj}}^n)=0$. As a result, we obtain the following matrix equation instead of (8)

$$\frac{1}{n}\sum_{q=1}^{Q}\left[\sum_{i=1}^{n}\frac{v_q(t_i^n)}{t_i^n-t_{0j}^n}+\sum_{i=1}^{n}K_{pq}\left(t_i^n,t_{0j}^n\right)v_q\left(t_i^n\right)\right]=f_p\left(t_{0j}^n\right),\quad (j=1,2,\ldots n-1)$$

$$\frac{1}{n}\sum_{q=1}^{Q}\left[\sum_{i=1}^{n}M_{pq}\left(t_i^n\right)v_q\left(t_i^n\right)\right]=c_p\qquad p=1,2,\ldots Q. \quad (11)$$

E. Computing Radiation Patterns With MDS

The total far-field radiation pattern (RP) as a function of the viewing angle φ_0 is defined as follows:

$$\Phi(\varphi_0)=\Phi_Q+\Phi_0 \equiv \lim_{r_0\to\infty}\sqrt{kr_0}e^{-jkr_0}[U(r_0,\varphi_0)+U_0(r_0,\varphi_0)] \quad (12)$$

where $\Phi_0(\varphi_0)=e^{kb\cos(\varphi_0-\beta)}$ is RP of the CSP feed in free space. The scattered field RP for the E -case is obtained after using the asymptotic representation of the Hankel function and above-mentioned quadrature formulas of interpolation type

$$\Phi_Q(\varphi_0)\approx\frac{1}{n}\left(\frac{2\pi}{j}\right)^{1/2}\times\sum_{q=1}^{Q}\left[\sum_{i=1}^{n}e^{-jkr_q(t_i^n)\cos(\varphi_q(t_i^n)-\varphi_0)}v_q\left(t_i^n\right)\right]. \quad (13)$$

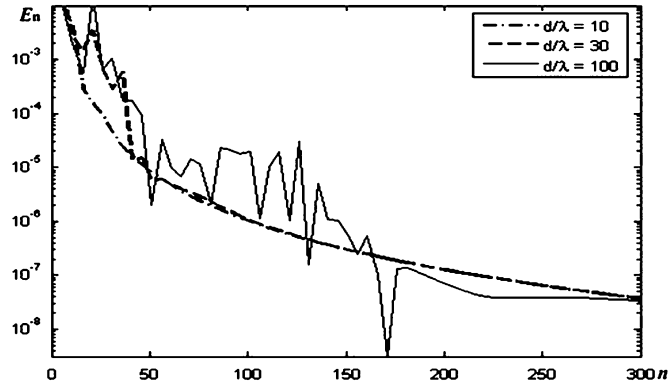

Fig. 2. MDS mean-square errors (E_n) in the radiation patterns of symmetric parabolic antennas having aperture sizes 100 λ (solid), 30 λ (dashed) and 10 λ (dash-dotted curve) versus the interpolation polynomial order n.

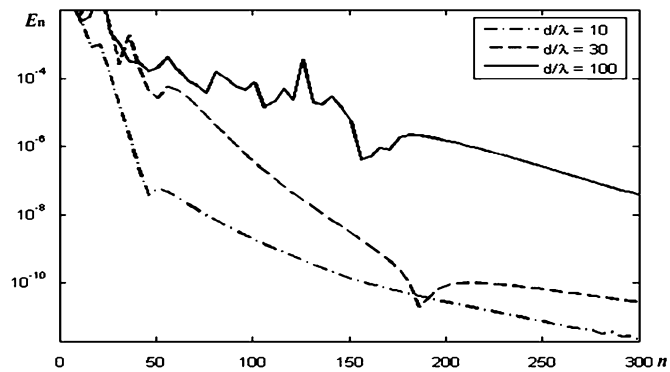

Fig. 3. MDS mean-square errors (E_n) in the surface current functions of the same symmetric parabolic antennas as in Fig. 2 versus the interpolation order n.

Here, $v_n(t_q^n)$ are the values of the current function in the Chebyshev nodes.

Given the pattern, the directivity of antenna is found as a ratio of the power radiated in the main beam direction $(\varphi_0=\varphi_{\max})$ to the average radiated power

$$D=\frac{2\pi}{P}|\Phi(\varphi_{\max})|^2,\quad P=\int_0^{2\pi}|\Phi(\varphi_0)|^2d\varphi_0. \quad (14)$$

Note, that the directivity of a CSP feed in free space (see [15]) is $D_0=e^{2kb}/I_0(2kb)$, where I_0 is the modified Bessel function.

F. Validation of Results

On solving (11), we obtain approximate solution to our SIEs in the form of interpolation polynomial, of the order n, for the unknown surface-current functions. In [17]–[19] it was proven that this MDS solution tends to exact ones, if $n\to\infty$, as $O(\kappa^{-n})$, where $\kappa\geq 2$ is the smoothness degree of $K_{pq}(t,t_0)$. Fig. 2 shows mean-square errors in radiation patterns of symmetric parabolic reflectors having 100λ, 30λ, and 10λ apertures, modeled with MDS (E-case), as a function of n.

To validate our numerical results, we have compared them with the data available from literature. In Fig. 4, we show normalized radiation patterns (NRP) of a symmetric parabolic reflector with $d=10\lambda$ and $f=0.5d$ illuminated by the E-type CSP-source with $kb=2.6$ located in the focus. The patterns

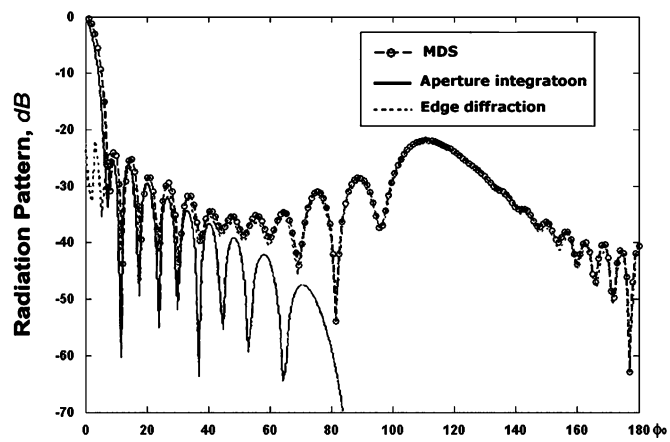

Fig. 4. NRPs of symmetric parabolic reflector fed by in-focus CSP ($kb = 2.6$), computed by different methods: PO&UTD [2], MAR [16] and MDS.

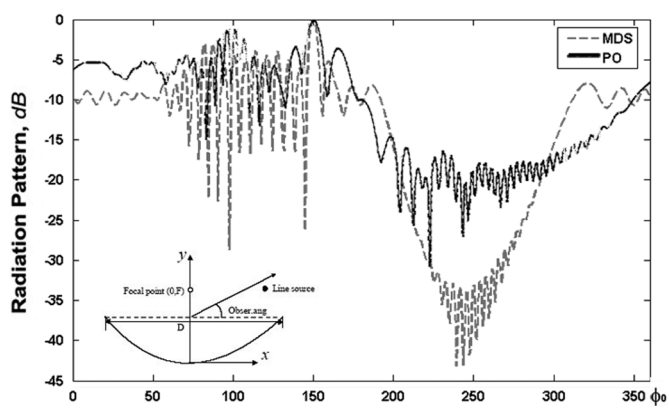

Fig. 5. NRPs of parabolic reflector fed by a defocused line current ($kb = 0$) computed by different methods. PO results are borrowed from [23].

were computed with PO and UTD [1], MAR [16] and our MDS. The latter two curves are undistinguishable and agree with PO and UTD in the main-beam and spillover regions, respectively.

In Fig. 5, we show NRPs for a parabolic reflector with offset line-current feed ($kb = 0$) computed with PO [24], MAR [16], and our MDS. The reflector parameters are $d = 15\lambda$ and $f = 4\lambda$, and the line source is located at $(5\lambda, 4\lambda)$.

Again MAR and MDS produce the same results, while PO yields correct main beam (in orientation and width) however is considerably less accurate off this region. This is clearly because the source sits near to one of the edges of reflector.

As for the time expenses, the computation of a map of 250×250 values of the near-field for a 100-λ reflector takes only 5 min on the 2 GHz Athlon with a 1 Gb RAM used here.

III. Numerical Analysis of Single Reflectors

A. Symmetric Parabolic Antennas

Consider a single parabolic reflector defined by the curve $L = \{(x, y) \in R^2 : x = y^2/4f, y_1 < y < y_2\}$, where $y_1 = -y_2$ (see Fig. 6). In computations, we assumed that the CSP feed aperture center was placed at the geometrical focus of parabola, i.e., at $(x_0, y_0) = (p/2, 0)$, and sees its center, i.e., $\beta = 180^o$.

NRPs for a 30-λ reflector and several values of the feed aperture parameter kb are shown in Fig. 7. One can see the narrow

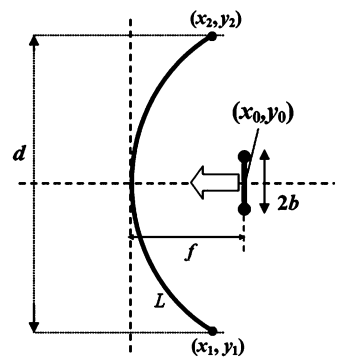

Fig. 6. Parabolic reflector symmetrically front-fed by a CSP feed.

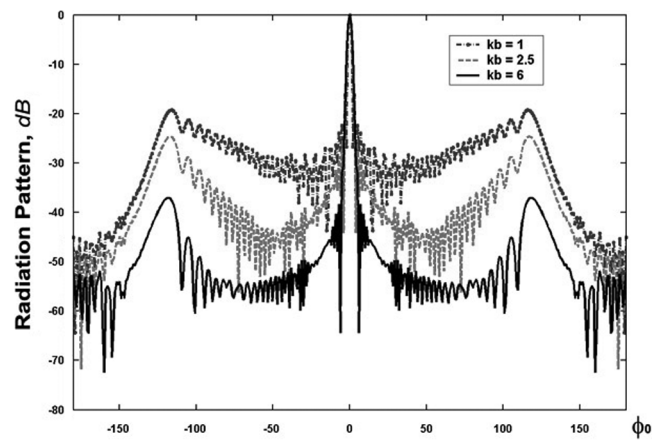

Fig. 7. NRPs of antenna shown in Fig. 3, for three different kb values and $d = 30\lambda, f = 0.5d$

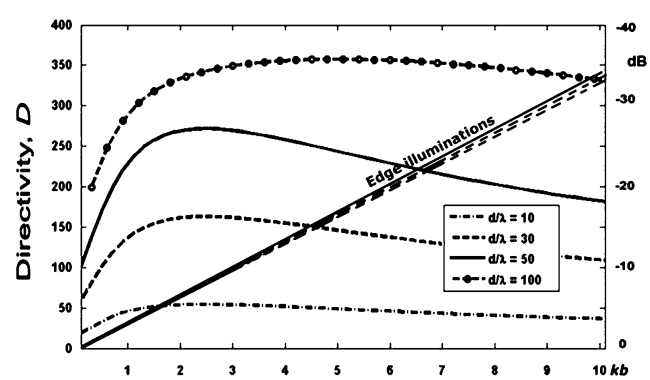

Fig. 8. Directivity and edge illumination of parabolic antennas with $f = 0.5d$ and different apertures as functions of kb.

main beam and two symmetric spillover lobes, which get lower if the incident beam is narrower. For all three feeds, the width of the main beam stays the same as it is determined primarily by the electrical size of reflector.

As common with quasioptical antennas, it is useful to introduce the edge illumination parameter, which is the ratio of the incident field value at the reflector edge to the field value at its center: $I^{\text{edge}} = 20\log(U_0^{\text{edge}}/U_0^{\text{cent}})$. This parameter values corresponding to the curves in Fig. 7 are $-6, -10$, and -21 dB.

Fig. 8 illustrates the directivities and edge illuminations, as a function of kb, for four symmetric parabolic antennas with the focal distance $f = 0.5d$ and the apertures from 10 to 100 λ.

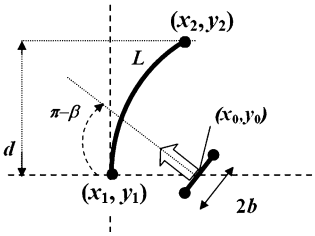

Fig. 9. 2-D offset parabolic reflector fed by CSP feed.

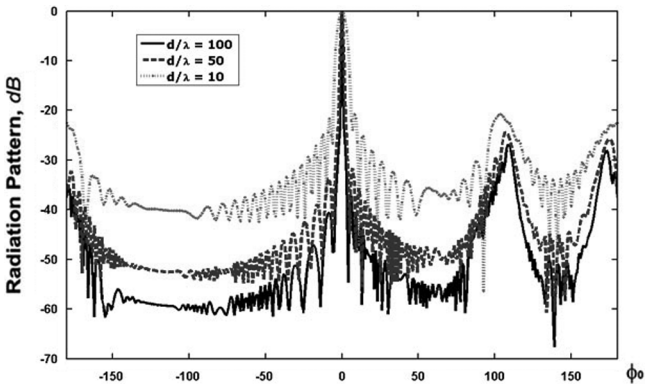

Fig. 10. NRPs of antenna shown in Fig. 9, for three values of d/λ and $kb = 9, \beta = 140°, f = 0.5a, y_2 = 0.5a, y_1 = 0.075a$, where a is the full parabolic arc size.

Antennas display maximum directivities at the kb values from 2.5 to 5, for the largest 100 λ antenna. This corresponds to the edge illumination gradually changing from -5 to -18 dB. Still in the latter case the difference for $kb > 2.5$ is insignificant and engineering rule of the -10 dB as an optimal edge illumination can be considered as good approximation.

B. Offset Parabolic Antennas

Consider now a non-symmetric parabolic reflector fed by in-focus CSP feed (Fig. 7). This geometry is attractive in practical design as it eliminates the aperture blockage by a real horn feed.

NRPs for several values of offset reflector size d/λ are shown in Fig. 10 for the feed with $kb = 9$ providing -10.5 dB edge illumination. Here $d = y_2 - y_1$, while $a = 2y_2$ is the aperture size of a full symmetric virtual reflector. Unlike a frontal illumination case, here one can see two non-symmetric spillover lobes. The level of the spillover lobes remains approximately the same as it is determined primarily by the edge illumination, i.e., kb.

Fig. 11(a) demonstrates the capabilities of the modeling of electrically large reflector antenna with MDS-based numerical algorithm. Here, offset reflector has the aperture of $d = 100\lambda$ that is clearly in the quasioptical domain. Accurate reconstruction of the near field reveals dramatic interference pattern of several types of waves. "Searchlight" main beam is well visible in the axial direction of parabola, while the other waves are apparently scattered by two differently illuminated edges, and the spillover field is also easily identified. "Quiet zone" of reflector is especially well observable after the visualization of the phase pattern of the total field—see Fig. 11(b).

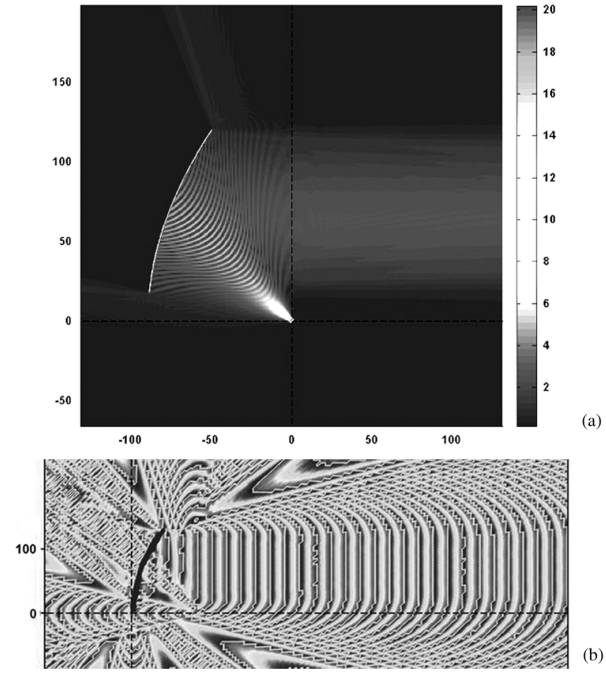

Fig. 11. (a) Near field intensity and (b) phase for an offset-parabolic reflector antenna, $d = 100\lambda, kb = 9, \beta = 140°, f = 0.5a, y_1 = 0.2a, y_2 = a/3$.

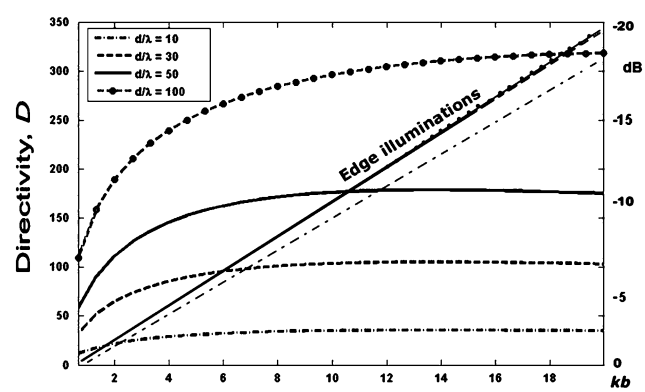

Fig. 12. Directivity and edge illumination of offset parabolic antennas with different apertures, as functions of kb; $\beta = 140°, f = 0.5a, y_1 = 0.2a, y_2 = a/3$.

This zone can be accurately quantified and contoured with MDS that should be useful for electromagnetic compatibility applications and in compact antenna testing range design.

The directivities and edge illuminations of offset parabolic antennas with $d = 10$ to 100 λ are shown in Fig. 12. The beam is directed at the antenna center so the edge illumination of both edges is kept equal. First three antennas display very broad maxima of directivities at kb around 14, which again corresponds to the edge illuminations of -10 dB, similar to symmetric parabolic antennas. The largest 100 λ antenna displays the maximum of the directivity at $kb = 20$ corresponding to the edge illumination of -18 dB.

IV. Numerical Analysis of Two-Reflector Systems

In such systems, separate scatterers are not so far from each other and thus may interact strongly. This places additional

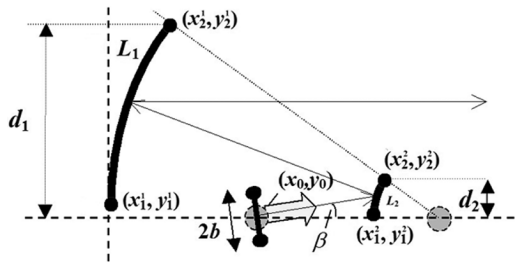

Fig. 13. Non-symmetric 2-D Cassegrain antenna fed by a CSP feed.

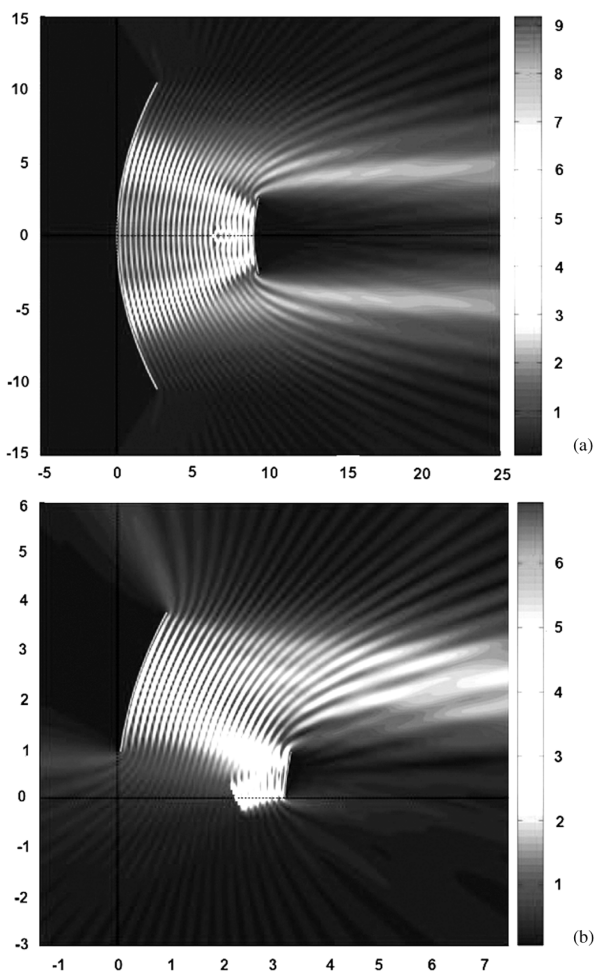

Fig. 14. Near fields of symmetric and non-symmetric Cassegrain antennas having (a) $d_1 = 30\lambda, d_2 = 7\lambda, kb = 9, \beta = 0$ and (b) $d_1 = 10\lambda, d_2 = 4\lambda, kb = 25, \beta = 22°, y_1^2 = 0, y_2^2 = y_1^1 = 0.12a, y_2^1 = 0.5a$.

burden to PO and GO-based simulation tools and may lead to inaccurate simulations.

A. Cassegrain Antennas

This system is used mainly to decrease the total size of antenna along the x-axis. A typical Cassegrain antenna has a main parabolic reflector and a feeding structure, which consists of a feed and a hyperbolic subreflector, to illuminate the main reflector (see Fig. 1). Non-symmetric Cassegrain designs (Fig. 13) are also of great interest as having more compact size along the

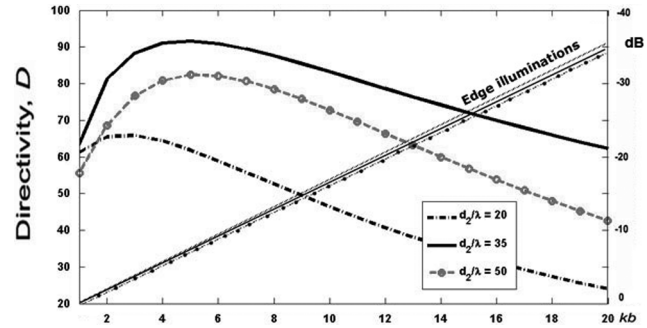

Fig. 15. Symmetric Cassegrain antenna directivity and subreflector edge illumination as functions of $kb; d_1 = 100\lambda, f = 0.5d$.

y-axis. In general case, the contours of two reflectors are characterized by the curves $L_1 = \{x = y^2/4f, y_1^1 < y < y_2^1\}$ and $L_2 = \{(x/H_1)^2 - (y/H_2)^2 = 1, y_1^2 < y < y_2^2\}$.

Two examples of the accurate visualization of the near fields of the Cassegrain systems using MDS (16) are shown in Fig. 14. Fig. 15 presents the dependences of the overall directivities of symmetric Cassegrain antennas and the subreflector edge illuminations on the CSP-feed aperture parameter, kb, for the same 100-λ main reflector and three different subreflectors of 20, 35, and 50-λ size.

Generally, there exists an optimal size of subreflector, around some 40λ in our case. This is because smaller subreflectors fail to illuminate full surface of main reflector, while larger subreflectors lead to reduced directivity values because of greater blockage of the main beams. The plots demonstrate that the optimal edge illumination of the optimal-size hyperbolic subreflector remains the same as for a single parabolic reflector of similar size (Fig. 4), i.e., -8 dB.

B. PACO Antennas

Axially symmetric design of parabolic subreflector and conical reflector (PACO) antenna is aimed at providing omnidirectional coverage with a horizontal or slightly tilted main beam, suited for the use in the base stations of the point-to-multipoint radio and TV links [25], [26]. We analyze numerically a 2-D analogue of the originally 3-D PACO antenna, in the E-polarization case (Fig. 16).

Therefore the contours of scatterers are $L_1 = \{y = -x^2/4f, y_1^1 < y < y_2^1\}$ for the parabolic reflector and $L_2 = \{y = -|x|^2, y_1^2 < y < y_2^2\}$ for the corner one except of the apex rounding of the 0.1 λ or smaller radius.

We have considered one CSP feed with $\beta = 90°$ (gray arrow) and two co-located CSP feeds with identical values of kb and $\beta = 90° \pm 30°$ (dotted arrows), elevated at 0.1λ above the apex. The latter model simulates a realistic TEM horn source whose radiation has minimum along the horn axis, to lower the diffraction by the cone apex and provide higher directivity. Figs. 17 and 18 show far and near fields for a $d = c = 30\lambda$ PACO antenna model with one and two feeds of $kb = 9.2$ that provides the maximum directivity in the double-feed case.

One can see a wider far-field main beam for the single-source fed PACO and correspondingly greater angular divergence of its near field, together with considerable scattering by the corner

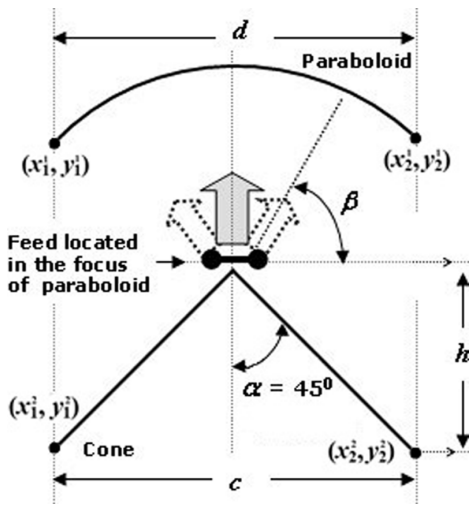

Fig. 16. 2-D model of a PACO antenna.

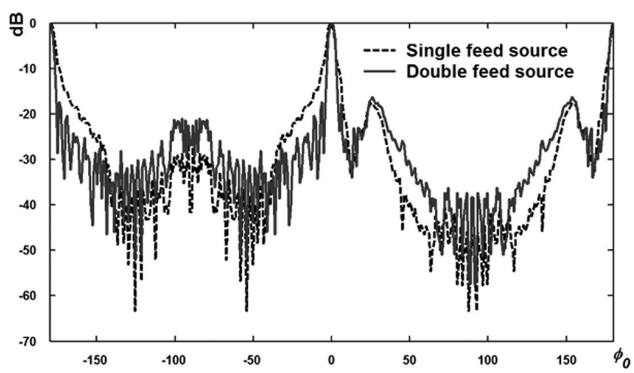

Fig. 17. NRPs for single and double-source PACO antennas with $d = c = 30\lambda$ and $kb = 9.2, f = 0.5d, x_0 = y_0 = 0$.

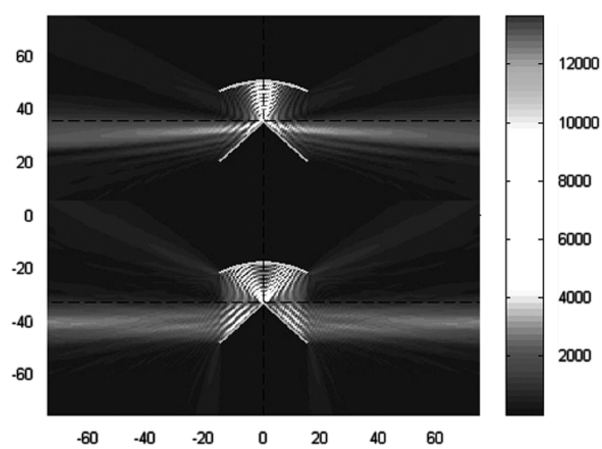

Fig. 18. Near fields for the same PACO antennas as in Fig. 17.

reflector (2-D "cone") apex. Besides, one can notice a better use of full "cone" surface in the case of double-feed arrangement.

Fig. 19 illustrates the directivities and the parabolic reflector edge illuminations as a function of CSP aperture size for single and two-source fed PACO antenna models with $d = c = 30\lambda$, i.e., with equal parabolic subreflector and corner reflector base sizes. It is visible that a single-feed design provides the peak directivity at $kb = 2.4$ however this quantity quickly degrades

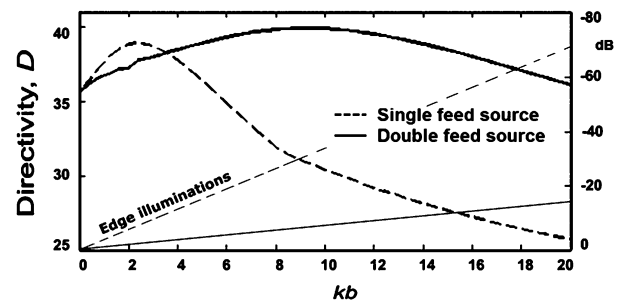

Fig. 19. Directivities and parabolic subreflector edge illuminations as functions of kb for single and double source PACO antennas having $d = c = 30\lambda$.

if the CSP beam is made sharper, because of the strong apex scattering. Double-feed one needs a narrower feed beam with $kb = 9.2$ to achieve optimum and yields somewhat higher peak value, which is much less sensitive to precise value of edge illumination. Generally, in either case the optimal edge illumination or parabolic subreflector is -12 to -15 dB. Note that the peak directivity of the 2-D PACO model is approximately four times lower than for the same 2-D parabolic reflector alone (compare to Fig. 8).

V. Conclusion

We have developed an economic and accurate numerical method able to compute 2-D models of electrically large multireflector antennas of arbitrary shape in seconds, with desired accuracy. This method, MDS, works with classical electric-field SIEs for the surface currents, however, unlike conventional MoM versions, has guaranteed and controlled convergence. It differs also from the analytical regularization methods [15], [16] which use the inversion of the static parts of these SIE. Unlike them, MDS exploits direct discretization of SIE, however in specific manner based on the Nystrom-type interpolation schemes of the highest accuracy that take account of the edge behavior of unknown function and the singular character of the kernel. So far, MDS has not been applied to the reflector analysis and, in our opinion, it has advantage over the other methods as either more accurate or more flexible and easier in implementation. MDS enables one to study near and far field patterns, surface currents, and also directivity and gain for various reflectors, feed locations, etc.

In this paper, we have presented MDS analysis for the case of E-polarization; the emphasis lies with the analysis of large one and two-reflector antennas that would be prohibited with a MoM subdomain discretization. As an example, we have performed systematic computations, which validate engineering rule-of-thumb of the -10 dB optimal edge illumination for single reflectors and show that it should be revised in favor of a stricter value in the case of two-reflector systems. The case of the H-polarization has also been considered, with a different Nystrom-type interpolation scheme, and the will be the subject of another publication As a future step, it is relatively straightforward to modify MDS for imperfect 2-D reflectors, either impedance or dielectric and resistive ones. Besides, it has a potential to be extended to the 3-D reflectors, at least those with rotational symmetry. Moreover, due to a lucky combination of high efficiency and controlled accuracy, the method can be efficiently used as a

core of the computer optimization software based on the gradient-type approach or on the global-search methods such as genetic and particle-swarm algorithms.

ACKNOWLEDGMENT

The authors wish to thank Prof. K. Schuenemann, Dr. T. Magath, and Prof. A. Altintas for many fruitful discussions, and Prof. M. Ando and Dr. T. Oguzer for comparison data used in Figs. 4 and 5.

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